Week 04: Replay Buffers & Target Networks

What Replay and Delayed Targets Change—and What They Do Not Prove

Reinforcement Learning & Robot Learning Core Curriculum

Presented by **DGIST ISL Team**

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Core Concepts: Deadly Triad · Temporal Decorrelation · Moving Targets · Multi-Seed Ablation

The Deadly Triad: Why Instability Can Appear

Function approximation, bootstrapping, and off-policy learning can interact poorly

1. Function Approx

Property: Parametric models $Q_\theta(s, a)$ share weights across states.

Risk: Updating Q at state s can shift predictions at distant states s'.

2. Bootstrapping

Property: Target $y = r + \gamma \max Q(s', \cdot)$ relies on existing guesses.

Risk: Errors in current predictions can feed into later targets.

3. Off-Policy Data

Property: Training distribution differs from greedy target policy.

Risk: The update distribution may not match the target policy, weakening standard convergence arguments.

Stabilizer 1: Circular Experience Replay Buffer

Spreading sample sets across stored time without claiming literally i.i.d. data

Sequential Window ($B = 32$)

- Transitions arrive sequentially: (s_t, a_t, r_t, s_{t+1}).
- **Mean all-pairs index gap:** 11.00.
- **Adjacent all-pair rate:** 6.2%.
- The window covers one contiguous timeline segment.

$s_0 \to s_1 \to s_2$

Uniform Replay Sampling

- Store N past transitions in FIFO buffer $\mathcal{D}$.
- Draw uniform minibatches: $\{e_i \sim \mathcal{D}\}_{i=1}^{B}$.
- **Mean all-pairs index gap:** 76.92.
- **Adjacent all-pair rate:** 0.6%.
- Spans stored time broadly; does not prove i.i.d.

e_{42} e_3 e_{18}

Stabilizer 2: Frozen Target Networks

Slowing one source of target motion with piecewise-frozen parameters

The Moving-Target Problem

Online-Only Target: $y = r + \gamma \max_{a'} Q_\theta(s', a')$

- An update to θ shifts prediction $Q_\theta(s, a)$ AND target y.
- Prediction and target can move after the same update.
- **Mean absolute target change: 0.014529**.

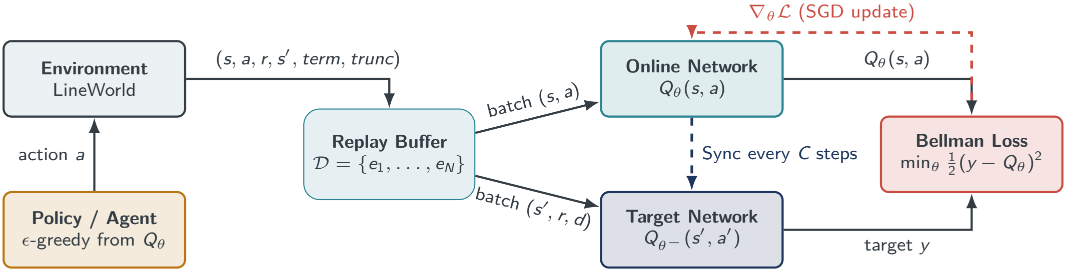
Frozen Target Parameterization

Target Network: $y = r + \gamma \max_{a'} Q_{\theta^-}(s', a')$

- Target weights θ^- remain frozen for C steps.
- Next-state parameters are fixed between synchronizations.
- **Mean absolute target change: 0.006766**.

System Architecture: The Stabilized DQN Loop

Interaction flow between environment, buffer, online network, and target network



Empirical Evidence: A Valid Null Result

Four linear-Q variants, 5 seeds, equal interaction/update budgets, fixed evaluation starts

Ablation Variant	Mean Return	Seed Std	Updates	Success Rate
Online Only	0.9850	0.0000	393	100%
Target Only	0.9850	0.0000	393	100%
Replay Only	0.9850	0.0000	393	100%
Replay + Target	0.9850	0.0000	393	100%

All rows: 400 interactions and 393 optimizer steps. Replay rows use $B = 8$; non-replay rows use $B = 1$, so compute is not matched.

Supported Claim

Under this exact LineWorld protocol, every configuration learned the same right-moving policy.

Unsupported Claim

The experiment does not rank the stabilizers. LineWorld is too easy to reveal a policy-level advantage.

Target Synchronization: Hard vs Soft Updates

Two standard paradigms for slowing target-parameter motion

Periodic Hard Updates (C steps)

- Copy weights abruptly every C optimizer steps:

$$\theta^- \leftarrow \theta \quad \text{every } C \text{ updates}$$

- **Advantage:** Target parameters are fixed between copies.
- **Tradeoff:** Targets can jump at synchronization.

Continuous Soft Updates (τ-Averaging)

- Interpolate weights continuously on every step:

$$\theta^- \leftarrow \tau\theta + (1-\tau)\theta^- \quad (\tau \approx 0.005)$$

- **Advantage:** Usually smoother target-parameter motion.
- Standard in continuous Actor-Critic methods (DDPG, SAC).

Failure Modes & Debugging Traps

Four major engineering bugs that destabilize DQN training

1. PREMATURE SAMPLING

Sampling without B stored items is undefined for sampling without replacement. Use an explicit warm-up rule.

2. TARGET OVER-SYNCHRONIZATION ($C = 1$)

Syncing after every optimizer step removes most of the intended delay; it does not by itself prove divergence.

3. BUFFER CAPACITY MISMATCH

Capacity trades off recency, coverage, memory, and stale data. There is no universally unstable size.

4. TERMINATION VS TRUNCATION

End rollouts on termination or truncation; mask bootstrap only for true termination.

Robotics Bridge: Off-Policy Sample Efficiency

Why replay buffers are the foundation of physical robot learning

The Physical Reality

- Physical execution is slower and costlier than model updates.
- Stored trials can support multiple optimizer steps.
- Report the update-to-data ratio and hardware interaction budget.

Real-to-Sim & Prioritized Buffers

- **Hybrid Buffers:** Mix real and simulated data deliberately, with proportions reported.
- **PER:** TD-error priorities are not identical to rarity, safety, or task importance.

Week 04 Quiz: Core Concepts & Mechanisms

Self-check: verify fundamental understanding of replay buffers & target networks

Q1. Replay and Policy Data

Why is arbitrary old replay easiest with off-policy learning?

A. Because on-policy methods cannot compute gradients.
B. Old data need not match the current on-policy distribution; reuse may require limits or correction.
C. Because buffers only support discrete actions.

Answer: B $\Longrightarrow$ Distribution mismatch matters.

Q2. Deadly Triad Components

Which three elements constitute the Deadly Triad?

A. Neural nets, Replay buffers, Discounting.
B. Function Approximation, Bootstrapping, and Off-Policy Learning.
C. Learning rate, Epsilon greedy, Mini-batches.

Answer: B $\Longrightarrow$ A known instability risk.

Week 04 Quiz: Buffer & Polyak Arithmetic

Verify hand-computations for replay indices and target update formulas

Problem 1: Circular Buffer Overwrite

Given: Buffer capacity $N = 4$. Slots 0, 1, 2, 3 are full. Current write pointer is at index 0.

Action: A new transition e_{new} is pushed.

Question: Which index is overwritten, and where is the pointer next?

Solution:

- Overwritten slot: **Index 0** (FIFO rule).
- Next pointer position: $(0 + 1) \bmod 4 = \mathbf{1}$.

Problem 2: Polyak Soft Update

Given: $\theta = 0.80, \theta^- = 0.40, \tau = 0.10$

Formula: $\theta^- \leftarrow \tau\theta + (1 - \tau)\theta^-$

Calculate: Updated target network parameter θ^-.

Solution:

- $\theta^- \leftarrow 0.10(0.80) + (1 - 0.10)(0.40)$
- $\theta^- = 0.08 + 0.36 = \mathbf{0.44}$
- Smooth continuous interpolation toward online θ.

Week 04 Quiz: Diagnostics & Practical Traps

Key engineering questions every RL practitioner must master

Q3. Setting $C = 1$ Degeneracy

What happens if target update frequency $C = 1$?
The target tracks online parameters almost immediately, removing most of the intended delay; divergence is not guaranteed.

Q4. Small Replay Buffer Risk

Does a buffer of size $N = 10$ always fail?
No. Capacity is a task-dependent tradeoff that must be measured.

Q5. Multi-Seed Evaluation Metric

Why report per-seed returns with mean and variance?
Means can hide failed runs, but variance alone cannot diagnose general stability.

Q6. Robot Sample Efficiency

Why is replay vital for physical hardware?
Hardware time is expensive; replay permits measured reuse of stored trials. Report the actual update-to-data ratio.